

POISSON COUNTS OF A MARKOVIAN RENEWAL PROCESS

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1. Introduction

Let F and G be two independent renewal processes commencing simultaneously at time $t = 0$. Let $F(x)$ be the distribution function (d.f.) of the intervals between successive renewals of the F -process and let $G(x)$ be the corresponding quantity for the G -process. If the renewals of the G -process occur at times $t = T_1, T_2, \dots, T_r, \dots$, we define the G -counts of the F -process by $P_{r,k}$ ($r = 1, 2, 3, \dots$; $k = 0, 1, 2, \dots$) where $P_{r,k}$ is the probability that the number of renewals of the F -process in the interval (T_{r-1}, T_r) is k . In particular, if the G -process is a Poisson process, $P_{r,k}$ are known as the Poisson counts of the F -renewal process. Kingman (1963) has derived the generating function of such Poisson counts. In this paper, this generating function is derived by a different method viz. that of using the distribution of the number of renewals in an arbitrary interval of time $(t_0, t_0 + t)$, where $t_0 \neq 0$ (see Cox (1962), p. 67). This method is then extended to obtain the Poisson counts of a Markov Renewal Process, as defined by Pyke (1961a), (1961b).

Let $T_r - T_{r-1} = x_r$, $T_0 = 0$ ($r = 1, 2, \dots$). Then T_{r-1} and x_r are independently distributed. Let $G_r(x)$ be the r -fold convolution of the d.f. $G(x)$. The distribution function of x_r is G and that of T_{r-1} is obviously G_{r-1} .

Let $A_k(t_0, t)$ be the probability that there are k renewals of the F -process in the interval $(t_0, t_0 + t)$ and let the corresponding probability generating function (p.g.f.) be

$$(1.1) \quad B(t, \xi, t_0) = \sum_{k=0}^{\infty} \xi^k A_k(t_0, t).$$

When $t_0 = 0$, we denote $B(t, \xi, 0)$ by $B(t, \xi)$ only. It can be readily seen that

$$(1.2) \quad P_{r,k} = \int_0^{\infty} \int_0^{\infty} A_k(t, x) dG_{r-1}(t) dG(x), \text{ for } r \geq 2, \quad \text{and}$$

$$(1.3) \quad P_{1,k} = \int_0^{\infty} A_k(0, x) dG(x).$$

The double generating function (with respect to r and k) of the G -counts of the F -process is, therefore,

$$(1.4) \quad \psi(\theta, \xi) = \sum_{r=1}^{\infty} \sum_{k=0}^{\infty} \theta^r \xi^k P_{r,k} \\ = \theta \int_0^{\infty} B(x, \xi) dG(x) + \sum_{r=2}^{\infty} \theta^r \int_0^{\infty} \int_0^{\infty} B(x, \xi, t) dG_{r-1}(t) dG(x).$$

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2. Poisson Counts of the F-process

The expression (1.4) for $\psi(\theta, \xi)$ can be evaluated explicitly, when the G-process is a Poisson process with parameter λ . For this we define the following Laplace transforms:

$$(2.1) \quad \int_0^\infty e^{-sx} dF(x) = f(s);$$

$$(2.2) \quad \int_0^\infty e^{-sx} B(x, \xi) dx = b(s, \xi);$$

$$(2.3) \quad \int_0^\infty e^{-sx} B(x, \xi, t_0) dx = b(s, \xi, t_0);$$

$$(2.4) \quad \int_0^\infty e^{-s_0 t_0} b(s, \xi, t_0) dt_0 = b^*(s, \xi, s_0).$$

Then it has been proved (see Cox (1962), pp. 67, 68, 37) that

$$(2.5) \quad b(s, \xi) = \frac{1 - f(s)}{s(1 - \xi f(s))};$$

$$(2.6) \quad b^*(s, \xi, s_0) = \frac{1}{ss_0} - \frac{(1 - \xi) \{f(s) - f(s_0)\}}{s(s_0 - s)(1 - \xi f(s))(1 - f(s_0))}.$$

When the G-process is a Poisson process,

$$(2.7) \quad G(x) = 1 - e^{-\lambda x}$$

and so (1.4) reduces to

$$(2.8) \quad \psi(\theta, \xi) = \theta \lambda b(\lambda, \xi) + \sum_{r=2}^{\infty} \lambda^2 \theta^2 \int_0^\infty \frac{(\lambda t \theta)^{r-2}}{(r-2)!} b(\lambda, \xi, t) e^{-\lambda t} dt$$

$$(2.9) \quad = \theta \lambda b(\lambda, \xi) + \theta^2 \lambda^2 b^*(\lambda, \xi, \lambda(1 - \theta))$$

$$(2.10) \quad = \frac{\theta}{1 - \theta} - \frac{\theta(1 - \xi)f(\lambda(1 - \theta))(1 - f(\lambda))}{(1 - \xi f(\lambda))(1 - f(\lambda(1 - \theta)))},$$

on account of (2.5) and (2.6). This agrees with Kingman's (1963) equation (16) on p. 1220, except for the fact that he takes $\lambda = 1$.

Incidentally, this shows that, if two renewal processes F_1 and F_2 have the same Poisson counts, expression (2.10) for F_1 will be identical with a similar expression for F_2 for all values of ξ and θ and this will show that the Laplace transforms of F_1 and F_2 are the same or that the two renewal processes are identical.

3. Poisson counts of a cumulative process

Suppose that a variable w , whose d.f. is $C(w)$ is associated with every renewal of the F-process. A cumulative process, then can be defined as

$$(3.1) \quad Z(t) = \sum_{i=1}^{N(t)} w_i$$

where as defined earlier, $N(t)$ is the number of renewals of the F-process in $(0, t)$. The w_i are all identically and independently distributed. Poisson counts of this cumulative process are obtained by considering the distribution of

$$(3.2) \quad Z_r = \sum w_j, \quad r = 1, 2, \dots$$

where the summation is over the renewals of the F-process that occur in (T_{r-1}, T_r) . If there are k such renewals (and the probability of this $P_{r,k}$), the Laplace transform of the distribution of Z_r , when k is fixed, is $\{c(s)\}^k$, where $c(s)$ is the Laplace transform of $C(w)$.

Hence the unconditional Laplace transform of the distribution of Z_r is

$$(3.3) \quad \sum_{k=0}^{\infty} [c(s)]^k P_{r,k}; \quad r = 1, 2, \dots$$

The generating function of these Laplace transforms is

$$(3.4) \quad \sum_{r=1}^{\infty} \sum_{k=0}^{\infty} \theta^r [c(s)]^k P_{r,k} = \psi(\theta, c(s)).$$

where $\psi(\theta, \xi)$ is given by (1.4).

4. Poisson counts of a Markovian renewal process

Markov Renewal Processes (M.R.P.) have been defined and studied extensively by Pyke (1961a), (1961b). We shall use the same notation as far as possible, as Pyke has used. A M.R.P. records at each time t , the number of times a system visits each of m states ($m < \infty$), in time t , if the transitions from state to state are according to a Markov chain and if the time required for each successive move is a random variable, the d.f. of which depends on the two states, between which the move is made. Let $[p_{ij}]$ be the transition probability matrix of the Markov chain, and let $F_{ij}(x)$ be the d.f. of the time taken to make a transition from state i to state j . We set $Q_{ij}(x) = p_{ij}F_{ij}(x)$, and define the Laplace-Stieltjes transform (L. - S.T.) as

$$(4.1) \quad q_{ij}(s) = \int_0^{\infty} e^{-sx} dQ_{ij}(x) \quad ; \quad i, j = 1, 2, \dots, m.$$

Let $q(s)$ be the $m \times m$ matrix of the $q_{ij}(s)$. We assume the system to be in state i initially and denote this by $Z_0 = i$; Z_t denotes the state of the system at time t . Further, we define the following quantities:

$$(4.2) \quad N_j(t) \quad = \quad \text{number of times the system visits state } j \text{ in the interval } (0, t) \quad j = 1, 2, \dots, m.$$

$$(4.3) \quad N_j(t_0, t) \quad = \quad \text{number of times the system visits state } j \text{ in the interval } (t_0, t_0 + t), \quad t_0 \neq 0.$$

$$(4.4) \quad \xi \quad = \quad \text{a diagonal matrix with elements } \xi_1, \dots, \xi_m.$$

$$(4.5) \quad \underline{e} \quad = \quad \text{a column vector } (m \times 1), \text{ with unit elements.}$$

$$(4.6) \quad B_i(t, \xi, t_0) = \sum_{n_1, \dots, n_m=0}^{\infty} \xi_1^{n_1} \xi_2^{n_2} \dots \xi_m^{n_m} \text{Prob}\{N_j(t_0, t) = n_j; \\ j = 1, 2, \dots, m | Z_0 = i\},$$

$$(4.7) \quad B_i(t, \xi) = \sum_{n_1, \dots, n_m=0}^{\infty} \xi_1^{n_1} \dots \xi_m^{n_m} \text{Prob}\{N_j(t) = n_j \mid Z_0 = i\}, \\ j = 1, 2, \dots, m$$

$$(4.8) \quad b_i(s, \xi, t_0) = L - S.T. \text{ of } B_i(t, \xi, t_0) \text{ with respect to } t \\ = \int_0^{\infty} e^{-st} d_t B_i(t, \xi, t_0),$$

$$(4.9) \quad b_i^*(s, \xi, s_0) = \int_0^{\infty} e^{-s_0 t_0} b_i(s, \xi, t_0) dt_0,$$

$$(4.10) \quad b_i(s, \xi) = \int_0^{\infty} e^{-st} d_t B_i(t, \xi).$$

$$(4.11) \quad \underline{b}^*(s, \xi, s_0) \text{ is the column vector of } b_i^*(s, \xi, s_0); i = 1, 2, \dots, m.$$

$$(4.12) \quad \underline{b}(s, \xi) \text{ is the column vector of } b_i(s, \xi); i = 1, 2, \dots, m.$$

The author has shown elsewhere (cf. Kshirsagar and Gupta, 1969a) that

$$(4.13) \quad \underline{b}^*(s, \xi, s_0) = \frac{1}{s_0} \underline{e} + \frac{1}{s_0 - s} (I - q(s_0))^{-1} (q(s) - q(s_0)) (\xi \underline{b}(s, \xi) - \underline{e})$$

and

$$(4.14) \quad \underline{b}(s, \xi) = (I - q(s)\xi)^{-1} (I - q(s)\underline{e}).$$

The last result has been proved in Kshirsagar and Gupta, (1969b); see also Cinlar (1968).

Poisson counts of the M.R.P. can now be defined by

$$(4.15) \quad P_{ir}(n_1, n_2, \dots, n_m) = \text{Prob}[N_j(T_{r-1}, R_r) = n_j | Z_0 = i] \quad j = 1, 2, \dots, m.$$

$$i = 1, 2, \dots, m$$

$$r = 1, 2, \dots$$

This is the joint distribution of the number of visits of the M.R.P. to the various states, in the random interval (T_{r-1}, T_r) defined for the Poisson process. Proceeding exactly in the same way, as in Section 2, the generating function of these counts is

$$\begin{aligned}
 (4.16) \quad \psi_1(\theta, \xi) &= \sum_{r=1}^{\infty} \theta^r \sum_{n_1, \dots, n_m=0}^{\infty} \xi_1^{n_1} \xi_2^{n_2} \dots \xi_m^{n_m} P_{ir}(n_1, n_2, \dots, n_m) \\
 &= \theta^2 \lambda^2 \int_0^{\infty} \int_0^{\infty} B_i(x, \xi, t) e^{-\lambda(1-\theta)t} e^{-\lambda x} dt dx \\
 &+ \theta \lambda \int_0^{\infty} B_i(x, \xi) e^{-\lambda x} dx \\
 &= (\theta \lambda) \frac{b_i(\lambda, \xi)}{\lambda} + (\theta^2 \lambda^2) \int_0^{\infty} \frac{b_i(\lambda, \xi, t)}{\lambda} e^{-\lambda(1-\theta)t} dt \\
 &= \theta b_i(\lambda, \xi) + \theta^2 \lambda b_i^*(\lambda, \xi, \lambda(1-\theta))
 \end{aligned}$$

We shall denote the column vector of the m generating functions $\psi_i(\theta, \xi)$, ($i = 1, 2, \dots, m$) by $\underline{\psi}(\theta, \xi)$ and on account of (4.16) and (4.13), it can be expressed as

$$\begin{aligned}
 (4.17) \quad \frac{\theta^2}{1-\theta} \underline{e} + \theta (I - q(\lambda)\xi)^{-1} (I - q(\lambda)) \underline{e} \\
 - \theta (I - q(\mu))^{-1} (q(\lambda) - q(\mu)) \xi (I - q(\lambda)\xi)^{-1} (I - \xi^{-1}) \underline{e}
 \end{aligned}$$

where $\mu = \lambda(1-\theta)$.

5. Cumulative process

Corresponding to each state visited, let there be a random variable, whose d.f. depends on the state and assume that these variables are independently distributed. Let w be the variable and its d.f. be $C_j(w)$, corresponding to the state j , ($j = 1, 2, \dots, m$). Then this w defines a cumulative process associated with the M.R.P. and is defined by

$$(5.1) \quad Z(t) = \sum_{\substack{\text{all transitions of} \\ \text{the M.R.P. in } (0, t)}} w$$

Poisson counts of this cumulative process can be defined by

$$(5.2) \quad Z_r = \sum w, \quad r = 1, 2, \dots$$

where the summation is over all transitions in the random interval (T_{r-1}, T_r) . If the system visits state j , n_j times in (T_{r-1}, T_r) , ($j = 1, \dots, m$), then conditional on $n_1, \dots, n_m$, the L.S.T. of Z_r is

$$(5.3) \quad \prod_{j=1}^m \{c_j(s)\}^{n_j}$$

where $c_j(s)$ is the L.-S.T. of $C_j(w)$, and hence the unconditional L.-S.T. is

$$(5.4) \quad \sum_{n_1, \dots, n_m=0}^{\infty} P_{ir}(n_1, \dots, n_m) \prod_{j=1}^m [c_j(s)]^{n_j}, \text{ if } Z_0 = i.$$

$r=1, 2, \dots, m.$

The generating function of these L.-S.T.'s is, therefore

$$(5.5) \quad \sum_{r=1}^m \theta^r \sum_{n_1, \dots, n_m=0}^{\infty} P_{ir}(n_1, \dots, n_m) \prod_{j=1}^m [c_j(s)]^{n_j} = \Psi_i(\theta, c(s)),$$

where ψ_i is given by (4.16) and (4.17) and $c(s)$ is the diagonal matrix of the m elements $c_j(s); j = 1, 2, \dots, m.$

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